

O2YMECH - Make-up/Midterm 2 - Solutions

① a) $\vec{a}_1 = (0, a)$ $\vec{a}_2 = (\frac{a}{2}, 0)$ $\vec{r}_1(0) = (1, 0)$
 $\vec{r}_2(0) = (0, 2)$ } initial conditions.

integrating $\vec{v}_1 = (0, at)$ $\vec{v}_2 = (\frac{at}{2}, 0)$

integrating $\vec{r}_1 = (1, \frac{at^2}{2})$ $\vec{r}_2 = (\frac{at^2}{4}, 2)$

$$\vec{v}_1 = (0, at) \quad \vec{v}_2 = (\frac{at}{2}, 0)$$

$$\vec{r}_1 = (1, \frac{at^2}{2}) \quad \vec{r}_2 = (\frac{at^2}{4}, 2)$$

$$\vec{F}_{net} = m\vec{a}_1 + m\vec{a}_2 = m(\vec{a}_1 + \vec{a}_2) = m(\frac{a}{2}, a)$$

$$\vec{N}_{net} = m\vec{r}_1 \times \vec{a}_1 + m\vec{r}_2 \times \vec{a}_2 = m(\vec{r}_1 \times \vec{a}_1 + \vec{r}_2 \times \vec{a}_2)$$

$$= m a \hat{k} = \frac{2ma\hat{k}}{2} = \vec{0}$$

b) $\vec{F}_{net} \neq 0 \Rightarrow$ Linear momentum is not conserved.

$\vec{N}_{net} = 0 \Rightarrow$ Angular momentum is conserved.

c) $T_{LAB} = \frac{1}{2} m v_1^2 + \frac{1}{2} m v_2^2 = \frac{m}{2} (v_1^2 + v_2^2) = \frac{m}{2} (a^2 t^2 + \frac{a^2 t^2}{4}) = \frac{ma^2 t^2}{2} \cdot \frac{5}{4} = \frac{5ma^2 t^2}{8}$

$T_{LAB} = \underbrace{T_{com}}_{\text{Kinetic energy of COM with respect to LAB frame}} + \underbrace{T'}_{\text{Kinetic energy in COM frame}} \Rightarrow T' = T_{LAB} - T_{com}$

$\frac{d}{dt} \vec{R}_{com} = \frac{1}{2m} (m\vec{v}_1 + m\vec{v}_2) = \frac{1}{2} (\frac{at}{2}, at)$

$\vec{v}_{com} = \frac{1}{2} (\frac{at}{2}, at) \Rightarrow T_{com} = \frac{1}{2} 2m v_{com}^2 = \frac{m}{4} (\frac{a^2 t^2}{4} + a^2 t^2) = \frac{m}{4} \frac{5a^2 t^2}{4} = \frac{5ma^2 t^2}{16}$

$$T' = \frac{5}{8} ma^2 t^2 - \frac{5}{16} ma^2 t^2 = \frac{5}{16} ma^2 t^2$$

② Amplitude $= x_0 e^{-\beta t} = x_0 e^{-\frac{\omega_0}{2Q} t} = x_0 e^{-\frac{\omega_0}{2Q} T_0 N} = x_0 e^{-\frac{\omega_0}{2Q} \frac{2\pi}{\omega_0} N} = x_0 e^{-\frac{\pi}{Q} N}$ we want this to be 1
 $N \approx \frac{Q}{\pi} \approx \frac{20}{3} = 30$ oscillations

③ a) $r = \frac{1}{a\theta} \Rightarrow \frac{1}{r} = a\theta \Rightarrow \frac{d}{d\theta} (\frac{1}{r}) = a \Rightarrow \frac{d^2}{d\theta^2} (\frac{1}{r}) = 0 \Rightarrow \frac{1}{r} = -\frac{\mu}{l^2} r^2 f(r) \Rightarrow \boxed{f(r) = -\frac{l^2}{\mu} \frac{1}{r^3}}$

b) $U(r) = - \int f(r) dr + C = + \frac{l^2}{\mu} \int \frac{dr}{r^3} + C = - \frac{l^2}{2\mu} \frac{1}{r^2} + C$

(4) a). Coriolis force is always in tangential direction. The wagon is constrained to move in radial direction. Therefore, Coriolis force will not cause any motion because of the rails.

- Centrifugal force $-m \vec{\Omega} \times (\vec{\Omega} \times \vec{r}) = m \Omega^2 r \hat{r}$ is always in radial direction. It causes a radial acceleration. Therefore, let us write the eq. of motion in radial direction:

$$F = m \ddot{r} \Rightarrow m \Omega^2 r = m \ddot{r} \Rightarrow \boxed{\ddot{r} - \Omega^2 r = 0}$$

b) $r(t) = e^{\lambda t}$
 $\dot{r}(t) = \lambda e^{\lambda t}$
 $\ddot{r}(t) = \lambda^2 e^{\lambda t}$

$\ddot{r} - \Omega^2 r = 0 \Rightarrow \lambda^2 e^{\lambda t} - \Omega^2 e^{\lambda t} = 0 \Rightarrow \lambda_{1,2} = \pm \Omega$: two solutions.

$r_1(t) = e^{\Omega t}$ and $r_2(t) = e^{-\Omega t}$

Apply Initial cond.

$$r(0) = A + B = r_0$$

$$\dot{r}(0) = A - B = 0 \Rightarrow A = B = \frac{r_0}{2}$$

$$\Rightarrow r(t) = \frac{r_0}{2} (e^{\Omega t} + e^{-\Omega t}) = r_0 \cosh(\Omega t)$$

General solution: $\vec{r}(t) = A r_1(t) + B r_2(t) = A e^{\Omega t} + B e^{-\Omega t}$

$$\dot{r}(t) = A \Omega e^{\Omega t} - B \Omega e^{-\Omega t}$$

c) $R = r_0 \cosh(\Omega t') \Rightarrow \frac{R}{r_0} = \cosh(\Omega t') \Rightarrow \Omega t' = \operatorname{arccosh}\left(\frac{R}{r_0}\right) \Rightarrow t' = \frac{1}{\Omega} \operatorname{arccosh}\left(\frac{R}{r_0}\right)$

- d) Centrifugal force is proportional to the distance of the wagon to the rotation axis; not its velocity (as in Coriolis force). Therefore, there is a force acting on the wagon in radial direction, which creates the acceleration.